EXECUTIVE CASE STUDY

PREDICTIVE MAINTENANCE

From rare-event prediction to governed maintenance decisions

Business question

Can routine machine operating conditions identify elevated failure risk early enough to support better maintenance decisions?

10,000	339	3.39%	7
observations	failures	failure prevalence	model configurations

Jupyter Notebook File: https://github.com/pateldev02/predictive-maintenance
Dataset File: https://archive.ics.uci.edu/dataset/601/ai4i+2020+predictive+maintenance+dataset

EXECUTIVE DECISION

1. Executive Summary

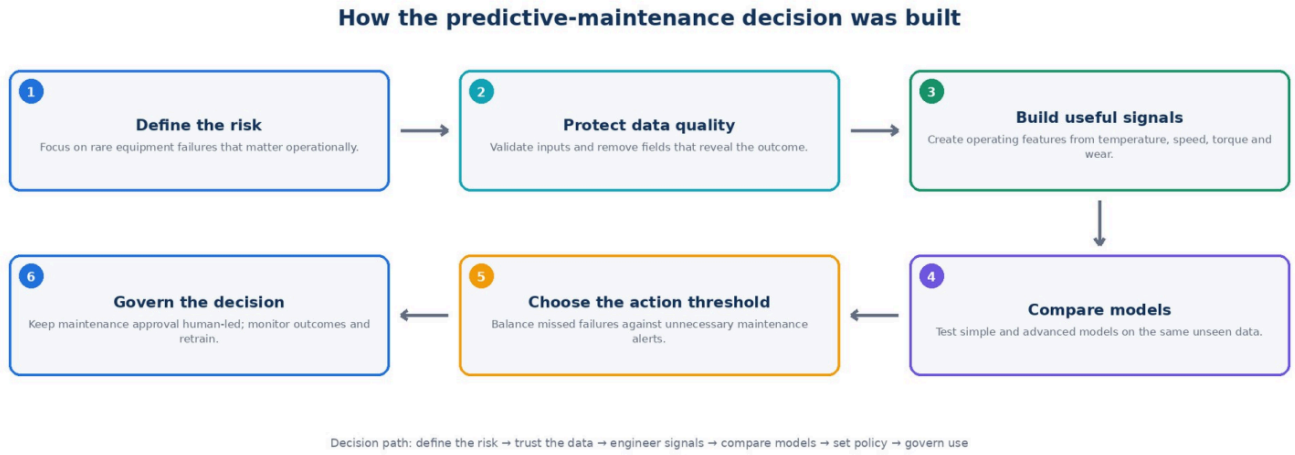
A concise view of the result, decision logic, and recommended enterprise path.

Recommendation

Use Random Forest as the leading pilot model and treat the decision threshold as a governed operating policy. At threshold 0.50 the model is highly precise; at 0.20 it detects more failures when the business prioritizes failure capture.

96.5%	80.9%	88.0%	89.1%
precision @ 0.50	recall @ 0.50	F1 @ 0.50	PR-AUC

Operational Requirement or Target: To achieve at least 85% detection rate of machine failures



What leadership should take away

- Accuracy was intentionally not used as the primary success measure: a naive model reached 96.6% accuracy while detecting zero failures.
- Random Forest produced the strongest overall balance at the default operating point and remained superior to XGBoost **under the 85%+ failure-detection scenario**.
- The preferred threshold depends on the economics of downtime, inspection capacity, asset criticality, and acceptable false-alarm burden.
- The proof of concept supports a controlled shadow-mode pilot, not autonomous maintenance decisions.

DECISION CONTEXT

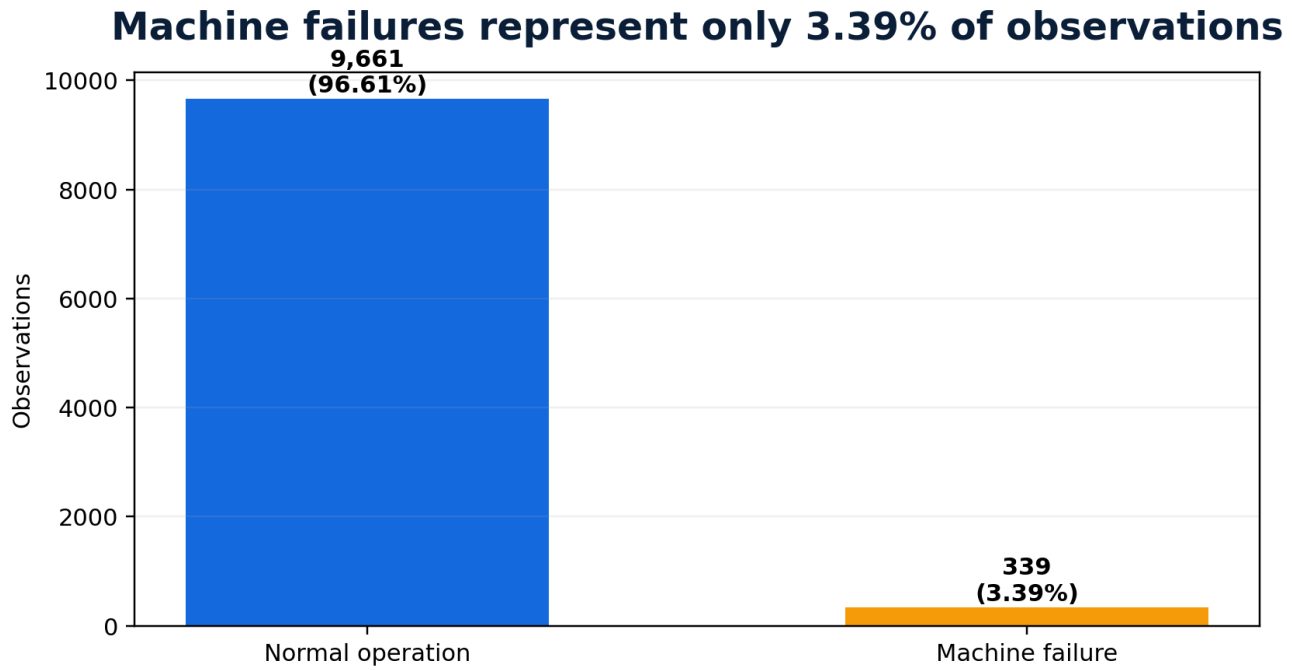
2. Predictive Maintenance as a Proactive Approach

The model is useful only if it improves decisions about uptime, maintenance capacity, and operational risk.

The operating challenge

Reactive maintenance exposes operations to unplanned downtime and downstream schedule disruption. Fixed preventive maintenance reduces some risk, but it can consume labor and parts on equipment that is not yet at elevated risk. Predictive maintenance adds a third option: prioritize attention using observed operating conditions.

The rare-event complication



Why 96.6% accuracy can be useless

Because only 3.39% of observations are failures, predicting “normal” every time achieves 96.6% accuracy and 0% failure recall. The project therefore focused on precision, recall, F1, ROC-AUC and PR-AUC.

Business interpretation of the core metrics

Metric	Plain-English meaning	Decision relevance
Precision	Of all failure alerts, how many were truly failures?	Controls unnecessary inspection burden.

Recall	Of all actual failures, how many did we catch?	Controls missed-failure risk.
F1	Balance between precision and recall.	Useful when both error types matter.
PR-AUC	Performance across precision-recall trade-offs.	Well suited to rare-event problems.
ROC-AUC	Ability to rank failures above normal states.	Useful for comparing discrimination across thresholds.

EVIDENCE FIRST

3. Data Quality, Leakage Control, and Operating Signal Discovery

The analysis established trust in the inputs before comparing algorithms.

0	0	5	3
missing values	duplicate rows	failure-mode flags excluded	engineered operating signals

Observed operating differences

Signal	Normal	Failure	Observed pattern
Torque	39.6 Nm mean	50.2 Nm mean	Higher load in failure observations
Tool wear	106.7 min mean	143.8 min mean	More accumulated wear
Rotational speed	1,507 rpm median	1,365 rpm median	Lower median RPM in failures
Type L failure rate	-	3.92%	Highest observed type-specific rate

Target leakage prevention

Critical modeling control

TWF, HDF, PWF, OSF and RNF were excluded from predictors because they directly describe failure outcomes. Including them would make the model look stronger than it could be in real use.

Feature engineering with operating meaning

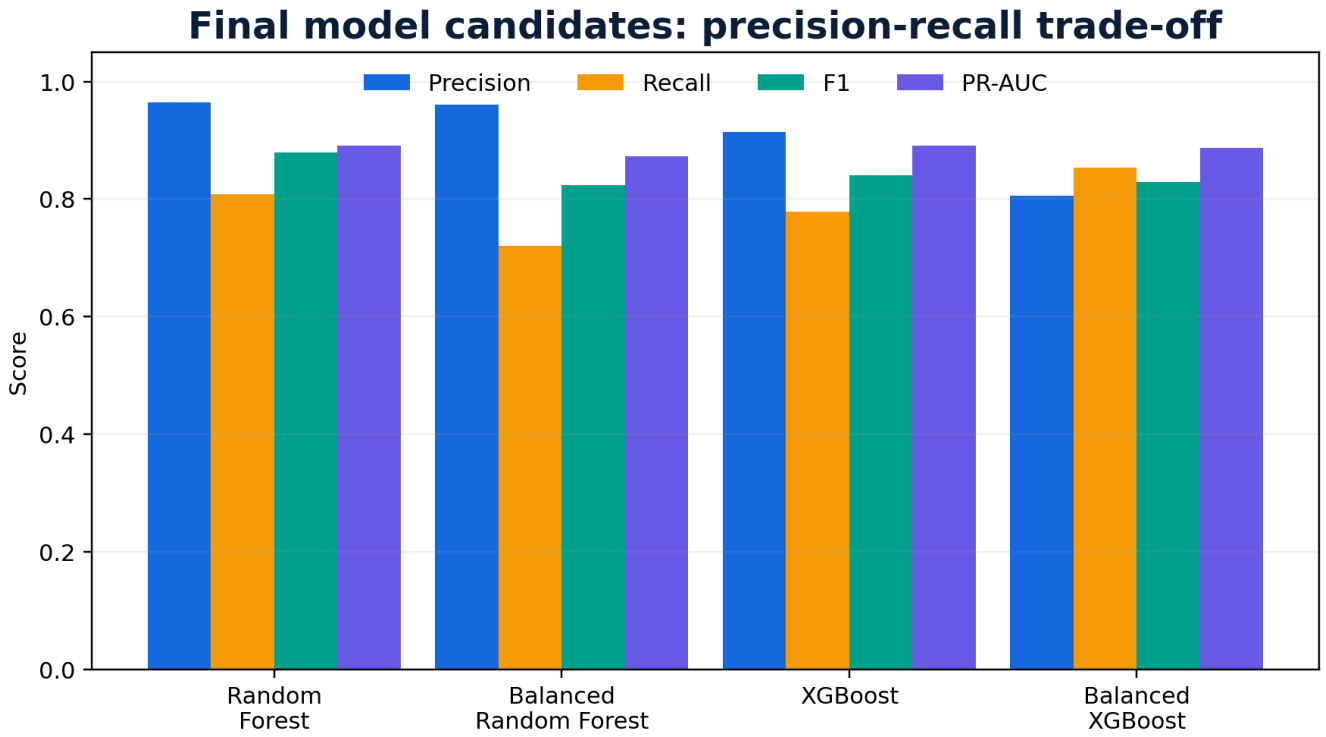
- Temperature difference = process temperature - air temperature.
- Mechanical power = torque × angular velocity.
- Tool-wear load = tool wear × torque.

- Engineered signals were evaluated empirically rather than assumed to improve performance.

MODEL TOURNAMENT

4. Model Selection Approach

Seven configurations were compared under a consistent train/test split and metric framework.



Model	Key result	Decision
Naive baseline	96.6% accuracy; 0% recall	Reject accuracy as a primary KPI
Logistic Regression	57.1% precision; 17.6% recall	Too conservative
Balanced Logistic	88.2% recall; 17.9% precision	Too many false alerts
Random Forest	96.5% precision; 80.9% recall; F1 88.0%	Leading candidate
Balanced RF	Recall fell to 72.1%	Balancing hurt performance
XGBoost	ROC-AUC 98.6%; PR-AUC 89.1%	Strong challenger
Balanced XGBoost	Recall 85.3%; precision 80.6%	Good high-recall option

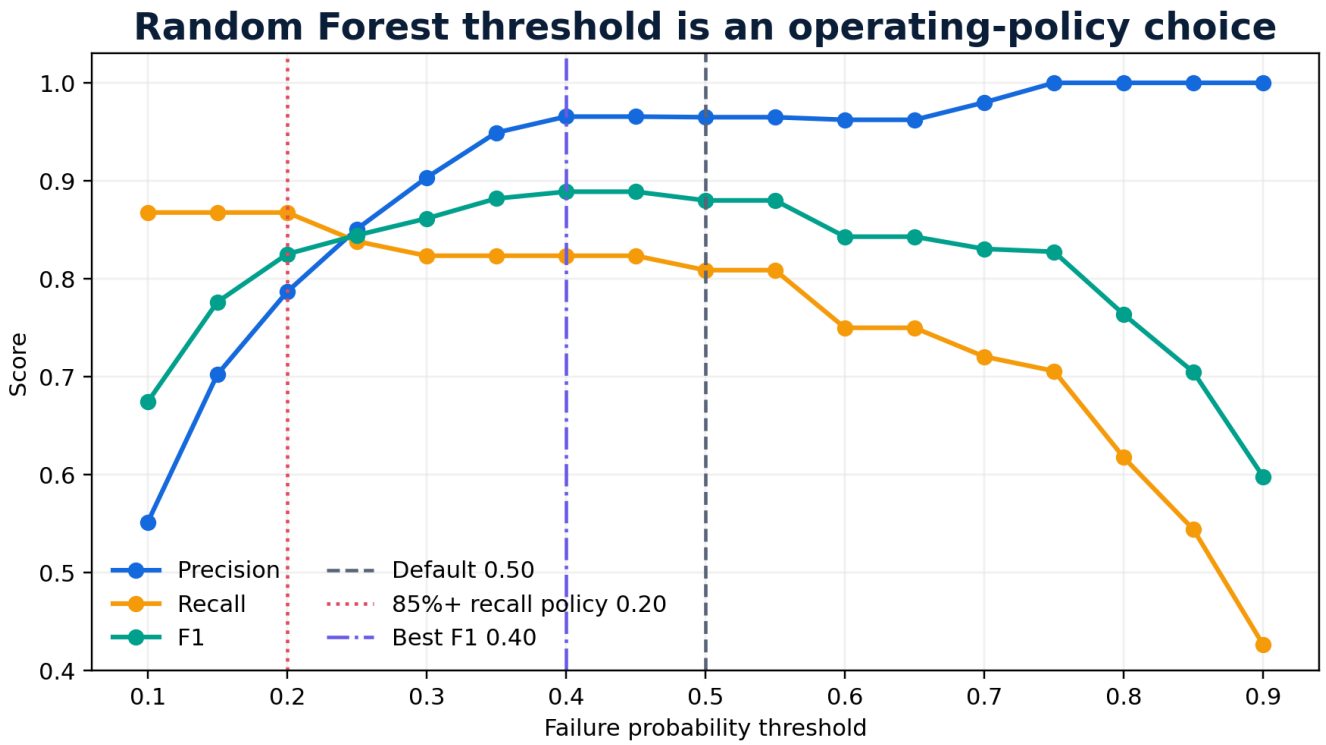
Key lesson

Class imbalance techniques were tested model by model. Balancing materially improved Logistic Regression recall but reduced Random Forest performance. The project used evidence rather than a one-size-fits-all rule.

OPERATING POLICY

5. Threshold Optimization: From Probability to Action

The same model can behave very differently depending on where the organization sets the alert threshold.



Policy	Threshold	Precision	Recall	F1	Predicted failures
Best F1	0.40	96.6%	82.4%	88.9%	-
Default / high confidence	0.50	96.5%	80.9%	88.0%	57
85%+ recall policy	0.20	78.7%	86.8%	82.5%	75

When both Random Forest and XGBoost were required to detect at least 85% of failures, Random Forest reached 86.8% recall at 78.7% precision. XGBoost reached the same recall only at a lower threshold and precision fell to 66.3%. Random Forest therefore delivered the same failure capture with fewer unnecessary alerts.

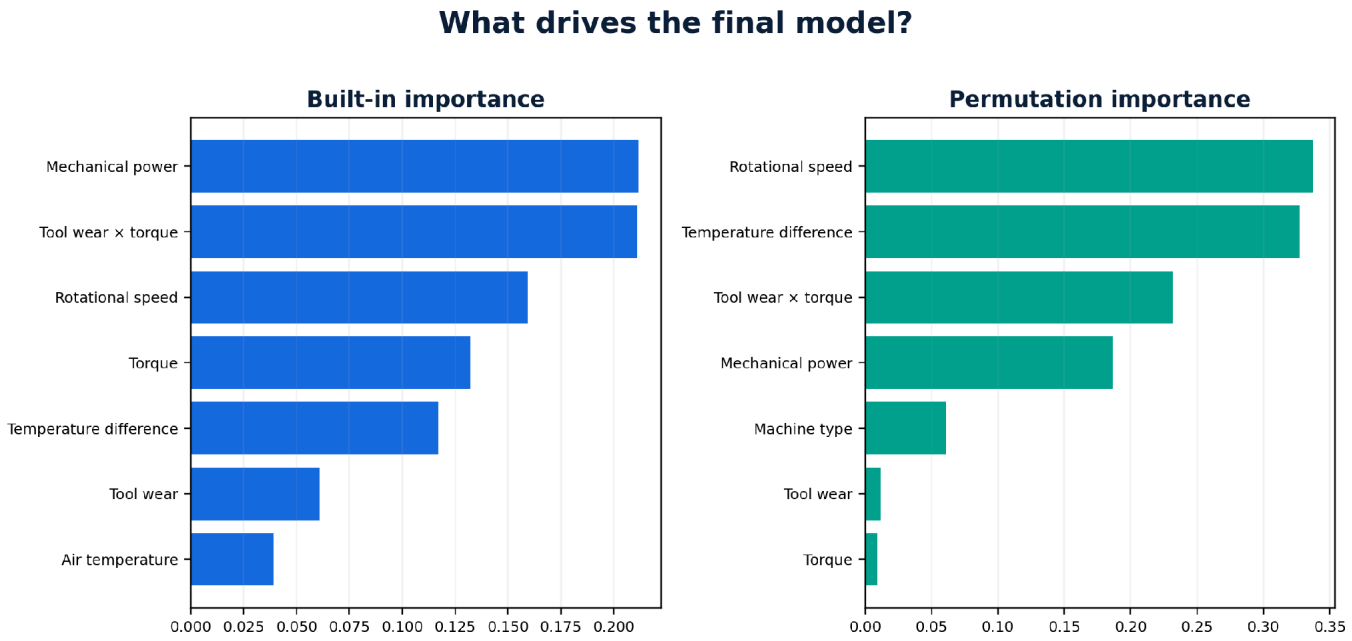
Governance point

The 85% recall target used here is a case study operating requirement, not an industry or regulatory standard. In production, the threshold should be approved against site specific downtime cost, maintenance capacity and asset criticality.

EXPLAINABILITY

6. What is Driving Predicted Risk?

Two importance methods were used to avoid relying on a single explanation technique.



Interpretation

The final model relies most strongly on mechanical power, tool-wear load, rotational speed, torque and temperature difference. Permutation importance independently highlighted rotational speed, temperature difference and tool-wear load as high-impact variables.

Important boundary

Predictive importance is not proof of causation. A high-importance variable can be a root cause, a symptom, or a correlated operating condition. Engineering and maintenance investigation remains necessary.

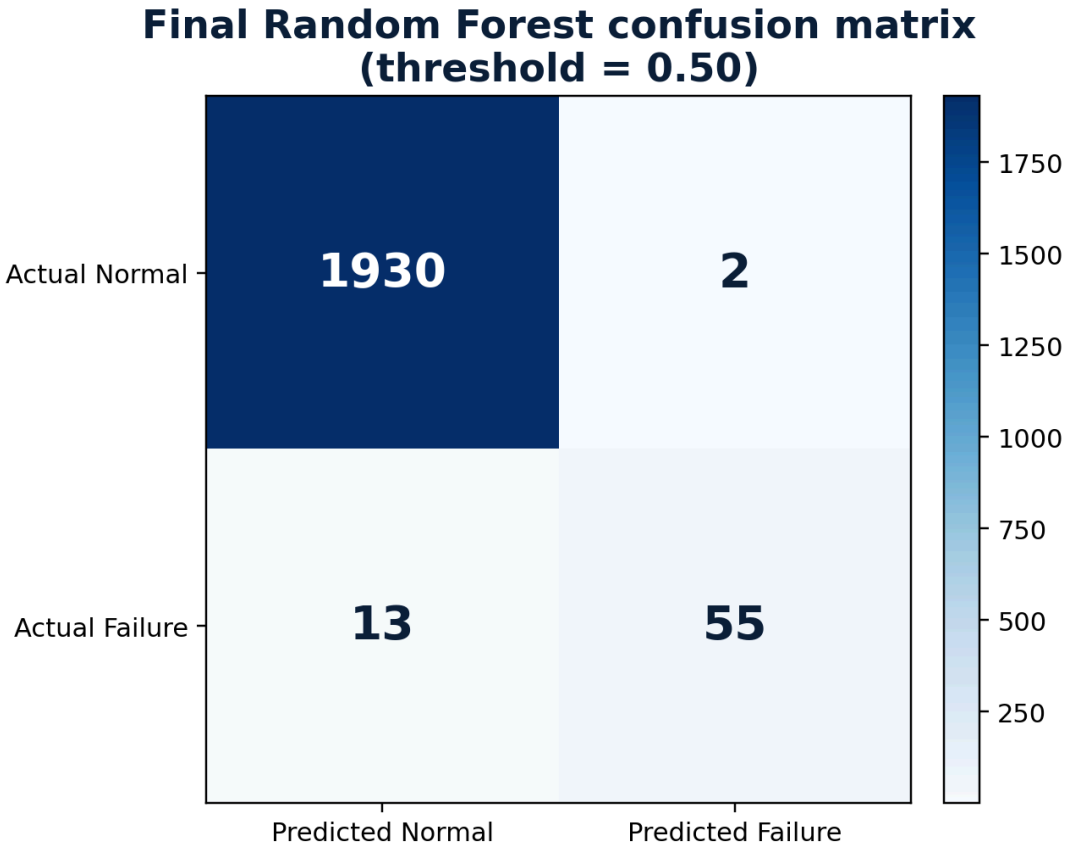
Risk to Action Plan Mapping

Risk tier	Illustrative score	Decision support
Low	<20%	Continue normal operation
Moderate	20%-49%	Increase monitoring / inspect context
High	50%-74%	Schedule maintenance inspection
Critical	≥75%	Immediate maintenance review

FINAL PERFORMANCE

7. Final Model Performance Evaluation and Assessment

The confusion matrix translates abstract metrics into operational counts on the held-out test set.



1,930	55	2	13
normal states correctly cleared	failures correctly detected	false alerts	missed failures

At threshold 0.50, the final Random Forest correctly identified 55 of 68 failures while generating only two false alerts among 1,932 normal observations. This operating point favors confidence and maintenance efficiency. A lower threshold captures four additional failures but produces more false alerts.

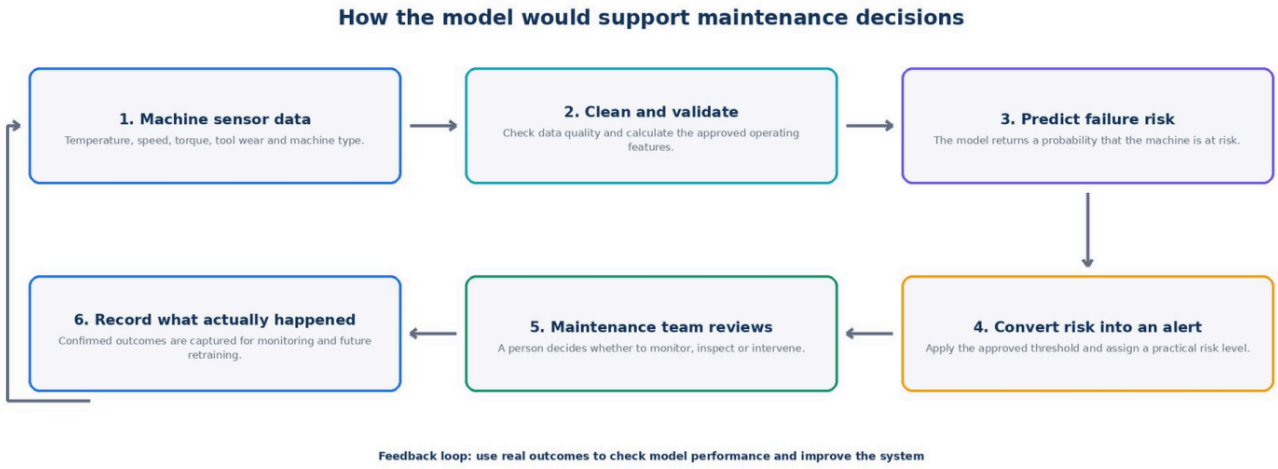
Where value could appear

Value lever	Potential business effect
Unplanned downtime	Earlier identification of higher-risk operating states
Maintenance productivity	Prioritize inspections where risk is highest
Spare parts / planning	Improve readiness around elevated-risk assets
Quality & reliability	Create a traceable evidence stream linking operating state, intervention and outcome
Capital planning	Build evidence on recurring high-risk equipment and failure mechanisms

CONTROL MODEL

8. An Approach to Model Evolution and Lifecycle in Enterprise

A prediction should enter a controlled process, not bypass it.



Minimum controls before production use

- Data integrity: equipment IDs, timestamps, sensor units, missingness, calibration status and source lineage must be controlled.
- Human-in-the-loop: model outputs should support, not replace, accountable maintenance and engineering decisions.
- Model governance: versioning, performance acceptance criteria, threshold history, approval and rollback should be documented.
- Monitoring: track drift, alert volume, false positives, missed failures and performance by asset/site/time period.
- Change control: retraining, feature changes and threshold changes should be assessed as controlled changes.

For regulated environments

If model output influences GMP decisions or validated equipment controls, intended use, validation, data integrity, auditability and change control requirements must be defined before deployment. This proof of concept does not establish that validated state.

SCALE-UP PLAN

9. From Proof of Concept to Enterprise Pilot

Move in gates so business value and control maturity increase together.

Gate	Purpose	Evidence required
1. Historical validation	Replay real plant history	Representative asset data; confirmed outcomes; stable features
2. Shadow mode	Score live data without changing maintenance decisions	Alert quality, latency, drift, workflow fit
3. Limited pilot	Use recommendations on selected assets with human approval	Defined owners, action SOP, benefit and error tracking
4. Controlled scale	Expand assets/sites after acceptance criteria are met	Repeatable controls, monitoring, support model, audit trail
5. Sustain	Monitor and revalidate	Drift thresholds, periodic review, retraining/change control

Business case inputs still required

- Cost per hour of unplanned downtime by critical asset.
- Cost of inspection / planned intervention and maintenance capacity.
- Failure severity and secondary impact by asset class.
- Sensor/integration costs and data-quality remediation effort.
- Value of additional production availability and reduced emergency work.

DECISION MEMO

10. Conclusion and Recommended Next Move

The model is promising; the next decision is whether the organization can validate it against real operations and economics.

Conclusion

Random Forest was selected because it delivered the strongest precision-recall balance at the default threshold and remained the better option when failure detection was increased to at least 85%. The project demonstrates a disciplined pathway from sensor data to a governed maintenance decision.

Recommended next actions

- Run a shadow-mode pilot on real telemetry for a representative set of critical assets.
- Link predictions to confirmed maintenance findings and actual failure outcomes.
- Quantify the economics of false alerts versus missed failures and approve an operating threshold.
- Validate performance by equipment type, site, operating regime and time period.
- Establish model ownership across operations, engineering, IT and quality before any decision automation.

Limitations

- The dataset is simulated/public and contains 339 failures; production performance may differ materially.
- Observations are operating snapshots rather than a true longitudinal degradation history for each machine.
- The test set comes from the same source distribution as training data and does not represent site-to-site transfer.
- Feature importance is predictive rather than causal.
- The selected thresholds are analytical operating examples, not validated safety or compliance limits.